

Vector Space

Rings:

Let R be a nonempty set with two binary operations, an operation of addition and an operation of multiplication. Then R is called a ring if the following axioms are satisfied:

[R1] For any $a, b, c \in R$, we have $(a + b) + c = a + (b + c)$.

[R2] There exists an element $0 \in R$, called the zero element, such that $a + 0 = 0 + a = a$ for every $a \in R$

[R3] For each $a \in R$ there exists an element $-a \in R$, called the negative of a , such that

$$a + (-a) = (-a) + a = 0$$

[R4] For any $a, b \in R$, we have $a + b = b + a$

[R5] For any $a, b, c \in R$, we have $(ab)c = a(bc)$.

[R6] For any $a, b, c \in R$, we have (i) $a(b + c) = ab + ac$, and (ii) $(b + c)a = ba + ca$.

Field: A commutative ring R with a unit element is called a field if every nonzero $a \in R$ has a multiplicative inverse; that is, there exists an element $a^{-1} \in R$ such that $aa^{-1} = a^{-1}a = 1$

Vector Space

Vector Space:

Let F be a field and suppose V is a set with a binary operation $+$ called addition assigning to any two elements $\mathbf{a}$ and $\mathbf{b}$ of V a unique sum $\mathbf{a} + \mathbf{b} \in V$. Suppose also that there is a second operation, called scalar multiplication, assigning to every $r \in F$ and $\mathbf{a} \in V$ a unique scalar multiple $r\mathbf{a} \in V$. Addition and scalar multiplication have the following properties.

- (1) Addition is commutative: $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ for all $\mathbf{a}, \mathbf{b} \in V$.
- (2) Addition is also associative: $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$ for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in V$.
- (3) V contains an additive identity $\mathbf{0}$: that is, $\mathbf{0} + \mathbf{a} = \mathbf{a}$ for all $\mathbf{a} \in V$.
- (4) For every element $\mathbf{v}$ of V , there is an element $-\mathbf{v}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Thus $-\mathbf{v}$ is an additive inverse of $\mathbf{v}$.

- (5) For all $\mathbf{a} \in V$, $1\mathbf{a} = \mathbf{a}$, where 1 is the multiplicative identity of F .
- (6) Scalar multiplication is associative: if $r, s \in F$ and $\mathbf{a} \in V$, then $(rs)\mathbf{a} = r(s\mathbf{a})$.
- (7) Scalar multiplication is distributive: if $r, s \in F$ and $\mathbf{a}, \mathbf{b} \in V$, then $r(\mathbf{a} + \mathbf{b}) = r\mathbf{a} + r\mathbf{b}$ and $(r + s)\mathbf{a} = r\mathbf{a} + s\mathbf{a}$.

Then V is called a *vector space over* F .

Let $V = \mathbb{R}^2$ and define addition and scalar multiplication operations as follows: If $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$, then define

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2)$$

and if k is any real number, then define

$$k\mathbf{u} = (ku_1, 0)$$

For example, if $u = (2, 4)$, $v = (-3, 5)$, and $k = 7$, then

$$\mathbf{u} + \mathbf{v} = (2 + (-3), 4 + 5) = (-1, 9)$$

$$k\mathbf{u} = 7\mathbf{u} = (7 \cdot 2, 0) = (14, 0)$$

The addition operation is the standard one from $\mathbb{R}^2$, but the scalar multiplication is not. In the exercises we will ask you to show that the first nine vector space axioms are satisfied. However, Axiom 10 fails to hold for certain vectors. For example, if $\mathbf{u} = (u_1, u_2)$ is such that $u_2 \neq 0$, then

$$1\mathbf{u} = 1(u_1, u_2) = (1 \cdot u_1, 0) = (u_1, 0) \neq \mathbf{u}$$

Thus, V is not a vector space with the stated operations.

Vector Space

Linearly Combination:

Let $v_1, \dots, v_k$ be in V with $a_1, \dots, a_k \in F$. Then the expression $a_1 v_1 + a_2 v_2 + \dots + a_n v_n$ is called the linear combination of the vectors $v_1, \dots, v_k$.

Linearly Independent and Dependent:

Let $v_1, \dots, v_k$ be in V . Then we say that $v_1, \dots, v_k$ are *linearly independent* (or simply *independent*) if the equation $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$, with $a_1, \dots, a_k \in F$, is satisfied only when $a_1 = a_2 = \dots = a_k = 0$. If the above equation has a nontrivial solution (i.e., some $a_i \neq 0$), we say that $v_1, \dots, v_k$ are *linearly dependent* (or simply *dependent*).

Spanning Sets:

Let $v_1, \dots, v_k$ be arbitrary elements of V . The *span* of $v_1, \dots, v_k$ is by definition the subset of V consisting of all linear combinations $\sum_{i=1}^k a_i v_i$

where $a_1, \dots, a_k$ are arbitrary elements of F . The span of $v_1, \dots, v_k$ is denoted by $\text{span} \{v_1, \dots, v_k\}$.

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Example1: Express $v = (3, 3, -4)$ in $\mathbb{R}^3$ as a linear combination of the vectors $u_1 = (1, 2, 3), u_2 = (2, 3, 7), u_3 = (3, 5, 6)$

Solution:

We seek scalars x, y, z such that $v = xu_1 + yu_2 + zu_3$; that is,

$$\begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix} = x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix} + z \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} \quad \text{or} \quad \begin{aligned} x + 2y + 3z &= 3 \\ 2x + 3y + 5z &= 7 \\ 3x + 7y + 6z &= -4 \end{aligned}$$

Reducing the system to echelon form yields

$$\begin{aligned} x + 2y + 3z &= 3 \\ -y - z &= 1 \\ y - 3z &= -13 \end{aligned} \quad \text{and then} \quad \begin{aligned} x + 2y + 3z &= 3 \\ -y - z &= 1 \\ -4z &= -12 \end{aligned}$$

Back-substitution yields the solution $x = 2, y = -4, z = 3$. Thus, $v = 2u_1 - 4u_2 + 3u_3$.

Vector Space

Example1: Express $v = (1, -2, 5)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, 1, 1), \quad u_2 = (1, 2, 3), \quad u_3 = (2, -1, 1)$$

Solution:

We seek scalars x, y, z , as yet unknown, such that $v = xu_1 + yu_2 + zu_3$. Thus, we require

$$\begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + z \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + y + 2z & = & 1 \\ x + 2y - z & = & -2 \\ x + 3y + z & = & 5 \end{array}$$

(For notational convenience, we write the vectors in $\mathbf{R}^3$ as columns, because it is then easier to find the equivalent system of linear equations.) Reducing the system to echelon form yields the triangular system

$$x + y + 2z = 1, \quad y - 3z = -3, \quad 5z = 10$$

The system is consistent and has a solution. Solving by back-substitution yields the solution $x = -6, y = 3, z = 2$. Thus, $v = -6u_1 + 3u_2 + 2u_3$.

Vector Space

Express $v = (2, -5, 3)$ in $\mathbf{R}^3$ as a linear combination of the vectors

$$u_1 = (1, -3, 2), \quad u_2 = (2, -4, -1), \quad u_3 = (1, -5, 7)$$

We seek scalars x, y, z , as yet unknown, such that $v = xu_1 + yu_2 + zu_3$. Thus, we require

$$\begin{bmatrix} 2 \\ -5 \\ 3 \end{bmatrix} = x \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} + y \begin{bmatrix} 2 \\ -4 \\ -1 \end{bmatrix} + z \begin{bmatrix} 1 \\ -5 \\ 7 \end{bmatrix} \quad \text{or} \quad \begin{array}{rcl} x + 2y + z & = & 2 \\ -3x - 4y - 5z & = & -5 \\ 2x - y + 7z & = & 3 \end{array}$$

Reducing the system to echelon form yields the system

$$x + 2y + z = 2, \quad 2y - 2z = 1, \quad 0 = 3$$

The system is inconsistent and so has no solution. Thus, v cannot be written as a linear combination of u_1, u_2, u_3 .

Vector Space

Express M as a linear combination of the matrices A, B, C , where

$$M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}, \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$$

Set M as a linear combination of A, B, C using unknown scalars x, y, z ; that is, set $M = xA + yB + zC$. This yields

$$\begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix} = x \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + y \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + z \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} x + y + z & x + 2y + z \\ x + 3y + 4z & x + 4y + 5z \end{bmatrix}$$

Form the equivalent system of equations by setting corresponding entries equal to each other:

$$x + y + z = 4, \quad x + 2y + z = 7, \quad x + 3y + 4z = 7, \quad x + 4y + 5z = 9$$

Reducing the system to echelon form yields

$$x + y + z = 4, \quad y = 3, \quad 3z = -3, \quad 4z = -4$$

The last equation drops out. Solving the system by back-substitution yields $z = -1, y = 3, x = 2$. Thus, $M = 2A + 3B - C$.

Vector Space

Show that the vectors $u_1 = (1, 1, 1)$, $u_2 = (1, 2, 3)$, $u_3 = (1, 5, 8)$ span $\mathbf{R}^3$.

We need to show that an arbitrary vector $v = (a, b, c)$ in $\mathbf{R}^3$ is a linear combination of u_1, u_2, u_3 . Set $v = xu_1 + yu_2 + zu_3$; that is, set

$$(a, b, c) = x(1, 1, 1) + y(1, 2, 3) + z(1, 5, 8) = (x + y + z, \quad x + 2y + 5z, \quad x + 3y + 8z)$$

Form the equivalent system and reduce it to echelon form:

$$\begin{array}{rcl} x + y + z = a & & x + y + z = a \\ x + 2y + 5z = b & \text{or} & y + 4z = b - a \\ x + 3y + 8z = c & & 2y + 7z = c - a \end{array} \quad \text{or} \quad \begin{array}{rcl} x + y + z = a & & x + y + z = a \\ y + 4z = b - a & & y + 4z = b - a \\ -z = c - 2b + a & & \end{array}$$

The above system is in echelon form and is consistent; in fact,

$$x = -a + 5b - 3c, \quad y = 3a - 7b + 4c, \quad z = a + 2b - c$$

is a solution. Thus, u_1, u_2, u_3 span $\mathbf{R}^3$.

Consider the vectors $\mathbf{u} = (1, 2, -1)$ and $\mathbf{v} = (6, 4, 2)$ in $\mathbb{R}^3$. Show that $\mathbf{w} = (9, 2, 7)$ is a linear combination of $\mathbf{u}$ and $\mathbf{v}$ and that $\mathbf{w}' = (4, -1, 8)$ is *not* a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

Solution In order for $\mathbf{w}$ to be a linear combination of $\mathbf{u}$ and $\mathbf{v}$, there must be scalars k_1 and k_2 such that $\mathbf{w} = k_1\mathbf{u} + k_2\mathbf{v}$; that is,

$$(9, 2, 7) = k_1(1, 2, -1) + k_2(6, 4, 2)$$

or

$$(9, 2, 7) = (k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

Equating corresponding components gives

$$\begin{aligned}k_1 + 6k_2 &= 9 \\2k_1 + 4k_2 &= 2 \\-k_1 + 2k_2 &= 7\end{aligned}$$

Solving this system using Gaussian elimination yields $k_1 = -3$, $k_2 = 2$, so

$$\mathbf{w} = -3\mathbf{u} + 2\mathbf{v}$$

Similarly, for $\mathbf{w}'$ to be a linear combination of $\mathbf{u}$ and $\mathbf{v}$, there must be scalars k_1 and k_2 such that $\mathbf{w}' = k_1\mathbf{u} + k_2\mathbf{v}$; that is,

$$(4, -1, 8) = k_1(1, 2, -1) + k_2(6, 4, 2)$$

or

$$(4, -1, 8) = (k_1 + 6k_2, 2k_1 + 4k_2, -k_1 + 2k_2)$$

Equating corresponding components gives

$$\begin{aligned}k_1 + 6k_2 &= 4 \\2k_1 + 4k_2 &= -1 \\-k_1 + 2k_2 &= 8\end{aligned}$$

This system of equations is inconsistent (verify), so no such scalars k_1 and k_2 exist. Consequently, $\mathbf{w}'$ is not a linear combination of $\mathbf{u}$ and $\mathbf{v}$.

Determine whether $\mathbf{v}_1 = (1, 1, 2)$, $\mathbf{v}_2 = (1, 0, 1)$, and $\mathbf{v}_3 = (2, 1, 3)$ span the vector space $\mathbb{R}^3$.

Solution We must determine whether an arbitrary vector $\mathbf{b} = (b_1, b_2, b_3)$ in $\mathbb{R}^3$ can be expressed as a linear combination

$$\mathbf{b} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3$$

of the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$. Expressing this equation in terms of components gives

$$(b_1, b_2, b_3) = k_1(1, 1, 2) + k_2(1, 0, 1) + k_3(2, 1, 3)$$

or

$$(b_1, b_2, b_3) = (k_1 + k_2 + 2k_3, k_1 + k_3, 2k_1 + k_2 + 3k_3)$$

or

$$\begin{aligned}k_1 + k_2 + 2k_3 &= b_1 \\k_1 + k_3 &= b_2 \\2k_1 + k_2 + 3k_3 &= b_3\end{aligned}$$

Thus, our problem reduces to ascertaining whether this system is consistent for all values of b_1 , b_2 , and b_3 . One way of doing this is to use parts (e) and (g) of Theorem 2.3.8, which state that the system is consistent if and only if its coefficient matrix

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

has a nonzero determinant. But this is *not* the case here; we leave it for you to confirm that $\det(A) = 0$, so $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$ do not span $\mathbb{R}^3$.

Vector Space

Spanning Sets

Let V be a vector space over K . Vectors $u_1, u_2, \dots, u_m$ in V are said to *span* V or to form a *spanning set* of V if every v in V is a linear combination of the vectors $u_1, u_2, \dots, u_m$ —that is, if there exist scalars $a_1, a_2, \dots, a_m$ in K such that

$$v = a_1u_1 + a_2u_2 + \cdots + a_mu_m$$

EXAMPLE 4.3 Consider the vector space $V = \mathbf{R}^3$.

(a) We claim that the following vectors form a spanning set of $\mathbf{R}^3$:

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then

$$v = ae_1 + be_2 + ce_3$$

For example, $v = (5, -6, 2) = -5e_1 - 6e_2 + 2e_3$.

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(b) We claim that the following vectors also form a spanning set of $\mathbf{R}^3$:

$$w_1 = (1, 1, 1), \quad w_2 = (1, 1, 0), \quad w_3 = (1, 0, 0)$$

Specifically, if $v = (a, b, c)$ is any vector in $\mathbf{R}^3$, then (Problem 4.62)

$$v = (a, b, c) = cw_1 + (b - c)w_2 + (a - b)w_3$$

For example, $v = (5, -6, 2) = 2w_1 - 8w_2 + 11w_3$.

(c) One can show (Problem 3.24) that $v = (2, 7, 8)$ cannot be written as a linear combination of the vectors

$$u_1 = (1, 2, 3), \quad u_2 = (1, 3, 5), \quad u_3 = (1, 5, 9)$$

Accordingly, u_1, u_2, u_3 do not span $\mathbf{R}^3$.

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DEFINITION: We say that the vectors $v_1, v_2, \dots, v_m$ in V are *linearly dependent* if there exist scalars $a_1, a_2, \dots, a_m$ in K , not all of them 0, such that

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$$

Otherwise, we say that the vectors are *linearly independent*.

(a) Let $u = (1, 1, 0)$, $v = (1, 3, 2)$, $w = (4, 9, 5)$. Then u , v , w are linearly dependent, because

$$3u + 5v - 2w = 3(1, 1, 0) + 5(1, 3, 2) - 2(4, 9, 5) = (0, 0, 0) = 0$$

(b) We show that the vectors $u = (1, 2, 3)$, $v = (2, 5, 7)$, $w = (1, 3, 5)$ are linearly independent. We form the vector equation $xu + yv + zw = 0$, where x, y, z are unknown scalars. This yields

$$x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + z \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ 2x + 5y + 3z = 0 \\ 3x + 7y + 5z = 0 \end{array} \quad \text{or} \quad \begin{array}{l} x + 2y + z = 0 \\ y + z = 0 \\ 2z = 0 \end{array}$$

Back-substitution yields $x = 0, y = 0, z = 0$. We have shown that

$$xu + yv + zw = 0 \quad \text{implies} \quad x = 0, \quad y = 0, \quad z = 0$$

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Example: Show that the space U generated by the vector $u_1 = (1, 2, -1, 3)$, $u_2 = (2, 4, 1, -2)$, and $u_3 = (3, 6, 3, -7)$ and the space V generated by the vectors $v_1 = (1, 2, -4, 11)$, and $v_2 = (2, 4, -5, 14)$ are equal; that is $U = V$.

Solution: Form the matrix A whose rows are the u_i , and row reduce A to row canonical form:

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{pmatrix} \xrightarrow[H_{31}(-3)]{H_{21}(-2)} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{pmatrix} \xrightarrow{H_{32}(-2)} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{H_2(\frac{1}{3})} \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 1 & -\frac{8}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{H_{12}(1)} \begin{pmatrix} 1 & 2 & 0 & \frac{1}{3} \\ 0 & 0 & 1 & -\frac{8}{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Now form the matrix B whose rows are the V_i , and row reduce B to row canonical form:

$$B = \begin{pmatrix} 1 & 2 & -4 & 11 \\ 2 & 4 & -5 & 14 \end{pmatrix} \xrightarrow{H_{21}(-2)} \begin{pmatrix} 1 & 2 & -4 & 11 \\ 0 & 0 & 3 & -8 \end{pmatrix} \xrightarrow{H_2(1/3)} \begin{pmatrix} 1 & 2 & -4 & 11 \\ 0 & 0 & 1 & -8/3 \end{pmatrix} \xrightarrow{H_{12}(4)} \begin{pmatrix} 1 & 2 & 0 & 1/3 \\ 0 & 0 & 1 & -8/3 \end{pmatrix}$$

Since the nonzero rows of the reduced matrices are identical, the row spaces A and B are equal and so $U = V$

Vector Space

Example: Let V be the set of ordered pairs of real numbers: $V = \{(a, b) : a, b \in \mathbb{R}\}$. Show that V is not a vector space over $\mathbb{R}$ with respect to each of the following operations of addition in V and scalar multiplication on V :

(i) $(a, b) + (c, d) = (a + c, b + d)$ and $k(a, b) = (ka, b)$;

(ii) $(a, b) + (c, d) = (a, b)$ and $k(a, b) = (ka, kb)$;

(iii) $(a, b) + (c, d) = (a + c, b + d)$ and $k(a, b) = (k^2a, k^2b)$;

Solution: In each case we will show one of the axioms of a vector space does not hold.

(i) Let $r = 1, s = 2, v = (3, 4)$. Then $(r + s)v = 3(3, 4) = (9, 4)$

$$rv + sv = 1(3, 4) + 2(3, 4) = (3, 4) + (6, 4) = (9, 8)$$

Since $(r + s)v \neq rv + sv$, axiom [7] does not hold.

(ii) Let $v = (1, 2), w = (3, 4)$. Then $v + w = (1, 2) + (3, 4) = (1, 2)$

$$w + v = (3, 4) + (1, 2) = (3, 4)$$

Since $v + w \neq w + v$, axiom [1] does not hold.

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(iii) Let $r = 1, s = 2, v = (3, 4)$. Then $(r + s)v = 3(3, 4) = (27, 36)$

$$rv + sv = 1(3, 4) + 2(3, 4) = (3, 4) + (12, 16) = (15, 20)$$

Since $(r + s)v \neq rv + sv$, axiom [7] does not hold.

Subspace: Let W be a subset of a vector space V over a field K . W is called a subspace of V if W is itself a vector space over K with respect to the operations of vector addition and scalar multiplication on V .

W is a subspace of V if and only if

(i) W is nonempty,

(ii) W is closed under vector addition: $v, w \in W$ implies $v + w \in W$,

(iii) W is closed under scalar multiplication: $v \in W$ implies $kv \in W$ for ever $k \in K$

Example: Let $V = \mathbb{R}^3$. Show that W is a subspace of V where:

(i) $W = \{(a, b, 0): a, b \in \mathbb{R}\}$, i.e. W is the xy-plane consisting of those vectors whose third component is 0;

(ii) $W = \{(a, b, c): a + b + c = 0\}$, i.e. W consist of those vectors each with the property that the sum of its component is zero.

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Solution: (i) $0 = (0, 0, 0) \in W$ since the third component of 0 is 0 . For any vector $v = (a, b, 0)$, $w = (c, d, 0)$ in W , and any scalars (real numbers) k and k' ,

$$kv + k'w = k(a, b, 0) + k'(c, d, 0) = (ka, kb, 0) + (k'c, k'd, 0) = (ka + k'c, kb + k'd, 0)$$

Thus $kv + k'w \in W$, and so W is a subspace of V .

(ii) $0 = (0, 0, 0) \in W$ since $0 + 0 + 0 = 0$. Suppose $v = (a, b, c)$, $w = (a', b', c')$ belong to W , i.e. $a + b + c = 0$ and $a' + b' + c' = 0$. Then for any scalars k and k' ,

$$\begin{aligned} kv + k'w &= k(a, b, c) + k'(a', b', c') = (ka, kb, kc) + (k'a', k'b', k'c') \\ &= (ka + k'a', kb + k'b', kc + k'c') \end{aligned}$$

$$\begin{aligned} \text{And furthermore, } (ka + k'a') + (kb + k'b') + (kc + k'c') &= k(a + b + c) + k'(a' + b' + c') \\ &= k \cdot 0 + k' \cdot 0 = 0 \end{aligned}$$

Thus $kv + k'w \in W$, and so W is a subspace of V .

Vector Space

Sum: Let U and W be subspaces of a vector space V . The sum of U and W , written, $U + W$, consists of all sums $u + w$ where $u \in U$ and $w \in W$:

$$U + W = \{u + w : u \in U, w \in W\}$$

Direct Sum: The vector space V is said to be the direct sum of its subspaces U and W , denoted by

$$V = U \oplus W$$

if every vector $v \in V$ can be written in one and only one way as $v = u + w$ where $u \in U$ and $w \in W$.

Example 1: In the vector space $\mathbb{R}^3$, let U be the xy -plane and W be the yz -plane:

$$U = \{(a, b, 0) : a, b \in \mathbb{R}\} \text{ and } W = \{(0, b, c) : b, c \in \mathbb{R}\}$$

Then $\mathbb{R}^3 = U + W$ since every vector in $\mathbb{R}^3$ is the sum of a vector in U and a vector in W . However, $\mathbb{R}^3$ is not direct sum of U and W since such sums are not unique; for example

$$(3, 5, 7) = (3, 1, 0) + (0, 4, 7) \text{ and also } (3, 5, 7) = (3, -4, 0) + (0, 9, 7)$$

The vector space V is the direct sum of its subspaces U and W if and only if

(i) $V = U + W$, and (ii) $U \cap W = \{0\}$

Suppose $V = U \oplus W$.

Then any $v \in V$ can be uniquely written in the form $v = u + w$ where $u \in U$ and $w \in W$. Thus, in particular, $V = U + W$.

Now suppose $v \in U \cap W$. Then (a) $v = v + 0$ where $v \in U$ and $0 \in W$ and (b) $v = 0 + v$ where $0 \in U$ and $v \in W$. Since such a sum for v must be unique, $v = 0$. Accordingly, $U \cap W = \{0\}$.

On the other hand, suppose $V = U + W$ and $U \cap W = \{0\}$

Let $v \in V$. Since $V = U + W$, there exists $u \in U$ and $w \in W$ such that $v = u + w$.

We need to show that such a sum is unique. Suppose also that $v = u' + w'$ where $u' \in U$ and $w' \in W$.

Then $u + w = u' + w'$ and so $u - u' = w' - w$. But $u - u' \in U$ and $w' - w \in W$; hence by $U \cap W = \{0\}$, $u - u' = 0$, $w' - w = 0$ and so $u = u'$, $w = w'$. Thus such a sum for $v \in V$ is unique and $V = U \oplus W$.

Basis and Dimension

Example 2: In $\mathbb{R}^3$, let U be the xy-plane and W be the z axis:

$$U = \{(a, b, 0) : a, b \in \mathbb{R}\} \text{ and } W = \{(0, 0, c) : c \in \mathbb{R}\}$$

Now any vector $(a, b, c) \in \mathbb{R}^3$ can be written as the sum of a vector in U and a vector in W in one and only one way: $(a, b, c) = (a, b, 0) + (0, 0, c)$

Accordingly, $\mathbb{R}^3$ is the direct sum of U and W , that is, $\mathbb{R}^3 = U \oplus W$

DEFINITION A: A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors is a basis of V if it has the following two properties:

(1) S is linearly independent. (2) S spans V .

DEFINITION B: A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors is a basis of V if every $v \in V$ can be written uniquely as a linear combination of the basis vectors.

Dimension: A vector space V is said to be of finite dimension n or n -dimensional, written $\dim V = n$ if V has a basis with n elements.

Example 1: Determine whether or not each of the following form a basis of $\mathbb{R}^3$: (a) $(1, 1, 1), (1, 0, 1)$; (b) $(1, 2, 3), (1, 3, 5), (1, 0, 1), (2, 3, 0)$; (c) $(1, 1, 1), (1, 2, 3), (2, -1, 1)$; (d) $(1, 1, 2), (1, 2, 5), (5, 3, 4)$.

Basis and Dimension

Solution: (a and b) No, because a basis of $\mathbb{R}^3$ must contain exactly three elements because $\dim R = 3$.

(c) The three vectors form a basis if and only if they are linearly independent. Thus, form the matrix whose rows are the given vectors, and row reduce the matrix to echelon form:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{pmatrix} \xrightarrow[H_{31}(-2)]{H_{21}(-1)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & -3 & -1 \end{pmatrix} \xrightarrow{H_{32}(3)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 5 \end{pmatrix}$$

The echelon matrix has no zero rows; hence, the three vectors are linearly independent, and so they do form a basis of $\mathbb{R}^3$.

(d) form the matrix whose rows are the given vectors, and row reduce the matrix to echelon form:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 5 \\ 5 & 3 & 4 \end{pmatrix} \xrightarrow[H_{31}(-5)]{H_{21}(-1)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & -2 & -6 \end{pmatrix} \xrightarrow{H_{32}(2)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

The echelon matrix has a zero row; hence, the three vectors are linearly dependent, and so they do not form a basis of $\mathbb{R}^3$.

Basis and Dimension

Example 2: Show that the vectors $v_1 = (1, 2, 1)$, $v_2 = (2, 9, 0)$, $v_3 = (3, 3, 4)$ form a basis of $\mathbb{R}^3$.

Solution: Form the matrix whose rows are the given vectors, and row reduce the matrix to echelon form:

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 9 & 0 \\ 3 & 3 & 4 \end{pmatrix} \xrightarrow[H_{31}(-3)]{H_{21}(-2)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & -2 \\ 0 & -3 & 1 \end{pmatrix} \xrightarrow{H_{32}(3/5)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & -2 \\ 0 & 0 & -1/5 \end{pmatrix}$$

The echelon matrix has no zero rows; hence, the three vectors are linearly independent, and so they do form a basis of $\mathbb{R}^3$.

Example 3: Let W be a space generated by the polynomials $v_1 = t^3 - 2t^2 + 4t + 1$, $v_2 = 2t^3 - 3t^2 + 9t - 1$, $v_3 = t^3 + 6t - 5$, $v_4 = 2t^3 - 5t^2 + 7t + 5$. Find a basis and the dimension of W .

Solution: The coordinate vectors of the given polynomials relative to the basis $\{t^3, t^2, t, 1\}$ are respectively $[v_1] = (1, -2, 4, 1)$, $[v_2] = (2, -3, 9, -1)$, $[v_3] = (1, 0, 6, -5)$, $[v_4] = (2, -5, 7, 5)$

Basis and Dimension

Form the matrix whose rows are the above coordinate vectors, and row reduce to echelon form

$$\begin{pmatrix} 1 & -2 & 4 & 1 \\ 2 & -3 & 9 & -1 \\ 1 & 0 & 6 & -5 \\ 2 & -5 & 7 & 5 \end{pmatrix} \xrightarrow{\substack{H_{21}(-2) \\ H_{31}(-1) \\ H_{41}(-2)}} \begin{pmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 2 & 2 & -6 \\ 0 & -1 & -1 & 3 \end{pmatrix} \xrightarrow{\substack{H_{32}(-2) \\ H_{42}(1)}} \begin{pmatrix} 1 & -2 & 4 & 1 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The nonzero rows $(1, -2, 4, 1)$ and $(0, 1, 1, -3)$ of the echelon matrix form a basis of the space generated by the coordinate vectors, and so the corresponding polynomials $t^3 - 2t^2 + 4t + 1$ and $t^2 + t - 3$ form a basis of W . Thus $\dim W = 2$.

Example 4: Find the dimension and a basis of the solution space W of the system

$$x + 2y + 2z - s + 3t = 0$$

$$x + 2y + 3z + s + t = 0$$

$$3x + 6y + 8z + s + 5t = 0$$

Basis and Dimension

Solution: Reduce the system to echelon form:

$$\begin{array}{l} x + 2y + 2z - s + 3t = 0 \\ z + 2s - 2t = 0 \\ 2z + 4s - 4t = 0 \end{array} \quad \longrightarrow \quad \begin{array}{l} x + 2y + 2z - s + 3t = 0 \\ z + 2s - 2t = 0 \end{array}$$

The system in echelon form has 2 (nonzero) equations in 5 unknowns; hence the dimension of the solution space W is $5 - 2 = 3$. the free variables are y , s and t .

Set (i) $y = 1, s = 0, t = 0$, we obtain $x = -2, z = 0$

(ii) $y = 0, s = 1, t = 0$, we obtain $x = 5, z = -2$

(iii) $y = 0, s = 0, t = 1$, we obtain $x = -7, z = 2$

Thus $v_1 = (-2, 1, 0, 0, 0)$, $v_2 = (5, 0, -2, 1, 0)$, $v_3 = (-7, 0, 2, 0, 1)$

The set $\{v_1, v_2, v_3\}$ is a basis of the solution space W .

Basis and Dimension

Problem 5: Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors $(1, -2, 5, -3)$, $(2, 3, 1, -4)$ and $(3, 8, -3, -5)$

(a) Find a basis and dimension of W . (b) Extend the basis of W to a basis of the whole space $\mathbb{R}^4$.

Solution:

(a) Form the matrix whose rows are the given vectors and reduce it to echelon form:

$$\begin{pmatrix} 1 & -2 & 5 & -3 \\ 2 & 3 & 1 & -4 \\ 3 & 8 & -3 & -5 \end{pmatrix} \xrightarrow[H_{31}(-3)]{H_{21}(-2)} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 14 & -18 & 4 \end{pmatrix} \xrightarrow{H_{32}(-2)} \begin{pmatrix} 1 & -2 & 5 & -3 \\ 0 & 7 & -9 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The nonzero rows $(1, -2, 5, -3)$ and $(0, 7, -9, 2)$ of the echelon matrix form a basis of the row space, that is, of W . Thus, in particular, $\dim W = 2$.

(b) We seek four linearly independent vectors, which include the above two vectors. The four vectors $(1, -2, 5, -3)$, $(0, 7, -9, 2)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$ are linearly independent (because they form an echelon matrix), and so they form a basis of $\mathbb{R}^4$, which is an extension of the basis of W .

Basis and Dimension

Problem 6: Let U and W be the following subspaces of $\mathbb{R}^4$: $U = \{(a, b, c, d): b + c + d = 0\}$, $W = \{(a, b, c, d): a + b = 0, c = 2d\}$. Find a basis and the dimension of (i) U (ii) W , (iii) $U \cap W$.

Solution:

(i) We seek a basis of the set of solutions (a, b, c, d) of the equation

$b + c + d = 0$ or $0 \cdot a + b + c + d = 0$. The free variables are a, c and d . Set

(a) $a = 1, c = 0, d = 0$ (b) $a = 0, c = 1, d = 0$ (c) $a = 0, c = 0, d = 1$ to obtain the respective solutions

$v_1 = (1, 0, 0, 0)$, $v_2 = (0, -1, 1, 0)$, $v_3 = (0, -1, 0, 1)$

The set $\{v_1, v_2, v_3\}$ is a basis of U , and $\dim U = 3$.

(ii) We seek a basis of the set of solutions (a, b, c, d) of the equation

$$\begin{array}{ccc} a + b = 0 & \longrightarrow & a + b = 0 \\ c = 2d & & c - 2d = 0 \end{array} \quad \text{The free variables are } b \text{ and } d. \text{ Set}$$

(a) $b = 1, d = 0$ (b) $b = 0, d = 1$ to obtain the respective solutions $v_1 = (-1, 1, 0, 0)$, $v_2 = (0, 0, 2, 1)$,

The set $\{v_1, v_2\}$ is a basis of W , and $\dim W = 2$.

Basis and Dimension

(iii) $U \cap W$ consists of those vectors (a, b, c, d) which satisfy the conditions defining U and the conditions defining W , i.e. the three equations

$$b + c + d = 0$$

$$a + d = 0$$

$$c = 2d$$



$$b + c + d = 0$$

$$a + d = 0$$

$$c - 2d = 0$$

The free variable is d . Set $d = 1$ to obtain the solution $v = (-1, -3, 2, 1)$

Thus $\{v\}$ is a basis of $U \cap W$, and $\dim(U \cap W) = 1$.

Linear Mappings

Mappings: Let A and B be arbitrary sets. Suppose to each $a \in A$ there is assigned a unique element of B ; the collection, f , of such assignments is called a function or mapping from A into B and is written $f: A \rightarrow B$.

One to One Mapping: A mapping $f: A \rightarrow B$ is said to one-to-one if different elements of A have distinct images; that is, if $a \neq a'$ implies $f(a) \neq f(a')$ or equivalently $f(a) \neq f(a')$ implies $a \neq a'$.

Onto Mapping: A mapping $f: A \rightarrow B$ is said to onto if ever $b \in B$ is the image of at least one $a \in A$.

Linear Mapping: Let V and U be vector space over the same field K . A mapping $F: V \rightarrow U$ is called a linear mapping if it satisfies the following two conditions:

- (i) For any $v, w \in V$, $F(v + w) = F(v) + F(w)$.
- (ii) For any $k \in K$ and any $v \in V$, $F(kv) = kF(v)$.

Kernel and Image of a linear mapping: Let $F: V \rightarrow U$ be a linear mapping. The image of F , written $\text{Im}F$, is the set of image points in U : $\text{Im}F = \{u \in U: F(v) = u \text{ for some } v \in V\}$

Linear Mappings

Kernel and Image of a linear mapping: Let $F: V \rightarrow U$ be a linear mapping. The image of F , written $\text{Im}F$, is the set of image points in U :

$$\text{Im}F = \{u \in U: F(v) = u \text{ for some } v \in V\}$$

The kernel of F , written $\text{Ker}F$, is the set of elements in V which map into $0 \in U$:

$$\text{Ker}F = \{v \in V: F(v) = 0\}$$

Show that the following mappings F are linear:

(i) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $F(x, y) = (x + y, x)$ (ii) $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $F(x, y, z) = 2x - 3y + 4z$

Solution: (i) Let $v = (a, b)$ and $w = (a', b')$; hence $v + w = (a + a', b + b')$ and $kv = (ka, kb)$, $k \in \mathbb{R}$

We have $F(v) = F(a, b) = (a + b, a)$ and $F(w) = F(a', b') = (a' + b', a')$

Thus, $F(v + w) = F(a + a', b + b') = (a + b + a' + b', a + a') = (a + b, a) + (a' + b', a') = F(v) + F(w)$

Linear Mappings

$$\text{and } F(kv) = F(ka, kb) = (ka + kb, ka) = k(a + b, a) = kF(v)$$

Since v , w and k were arbitrary, F is linear

(ii) Let $v = (a, b, c)$ and $w = (a', b', c')$; hence $v + w = (a + a', b + b', c + c')$ and $kv = (ka, kb, kc)$, $k \in \mathbb{R}$

$$\text{We have } F(v) = 2a - 3b + 4c \text{ and } F(w) = 2a' - 3b' + 4c'$$

$$\begin{aligned} \text{Thus, } F(v + w) &= F(a + a', b + b', c + c') = 2(a + a') - 3(b + b') + 4(c + c') \\ &= (2a - 3b + 4c) + (2a' - 3b' + 4c') = F(v) + F(w) \end{aligned}$$

$$\text{and } F(kv) = F(ka, kb, kc) = 2ka - 3kb + 4kc = k(2a - 3b + 4c) = kF(v)$$

Accordingly, F is linear

Linear Mappings

Show that the following mappings F are not linear:

(i) $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $F(x, y) = xy$ (ii) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $F(x, y) = (x + 1, 2y, x + y)$

(iii) $F: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $F(x, y, z) = (|x|, 0)$

Solution: (i) Let $v = (1, 2)$ and $w = (3, 4)$; hence $v + w = (4, 6)$

We have $F(v) = 1 \cdot 2 = 2$ and $F(w) = 3 \cdot 4 = 12$

Hence, $F(v + w) = F(4, 6) = 4 \cdot 6 = 24 \neq F(v) + F(w)$

Accordingly, F is not linear

(ii) Since $F(0, 0) = (1, 0, 0) \neq (0, 0, 0)$, F cannot be linear

(iii) Let $v = (1, 2, 3)$ and $k = -3$; hence $kv = (-3, -6, -9)$

We have $F(v) = (1, 0)$ and so $kF(v) = -3 \cdot (1, 0) = (-3, 0)$

$$F(kv) = F(-3, -6, -9) = (3, 0) \neq kF(v)$$

Hence, F is not linear

Linear Mappings

Let $F: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be the linear mapping defined by $F(x, y, s, t) = (x - y + s + t, x + 2s - t, x + y + 3s - 3t)$

Find a basis and the dimension of the (i) image U of F , (ii) kernel W of F .

Solution: (i) The images of the following generators of $\mathbb{R}^4$ generate the image U of F

$$F(1, 0, 0, 0) = (1, 1, 1) \quad F(0, 1, 0, 0) = (-1, 0, 1)$$

$$F(0, 0, 1, 0) = (1, 2, 3) \quad F(0, 0, 0, 1) = (1, -1, -3)$$

Form the matrix whose rows are the generators of U and row reduce to the echelon form

$$\begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & -1 & -3 \end{pmatrix} \xrightarrow[H_{41}(-1)]{H_{21}(1), H_{31}(-1)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & -2 & -4 \end{pmatrix} \xrightarrow{H_{42}(2)} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus $\{(1, 1, 1), (0, 1, 2)\}$ is a basis of U ; hence $\dim U = 2$

Linear Mappings

(ii) we seek the set of (x, y, s, t) such that $F(x, y, s, t) = (0, 0, 0)$, i.e.

$$F(x, y, s, t) = (x - y + s + t, x + 2s - t, x + y + 3s - 3t) = (0, 0, 0)$$

Set corresponding components equal to each other to form the following homogeneous system whose solution space is the kernel W of F :

$$\begin{array}{lcl} \begin{array}{l} x - y + s + t = 0 \\ x + 2s - t = 0 \\ x + y + 3s - 3t = 0 \end{array} & \xrightarrow{\quad} & \begin{array}{l} x - y + s + t = 0 \\ y + s - 2t = 0 \\ 2y + 2s - 4t = 0 \end{array} & \xrightarrow{\quad} & \begin{array}{l} x - y + s + t = 0 \\ y + s - 2t = 0 \end{array} \end{array}$$

The free variables are s and t ; hence $\dim W = 2$. Set

(a) $s = -1, t = 0$ to obtain the solution $(2, 1, -1, 0)$

(b) $s = 0, t = 1$ to obtain the solution $(1, 2, 0, 1)$

Thus $\{(2, 1, -1, 0), (1, 2, 0, 1)\}$ is a basis of W .

Linear Mappings

Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear mapping defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$

Find a basis and the dimension of the (i) image U of T , (ii) kernel W of T .

Solution: (i) The images of the following generators of $\mathbb{R}^3$ generate the image U of T

$$T(1, 0, 0) = (1, 0, 1) \quad T(0, 1, 0) = (2, 1, 1) \quad T(0, 0, 1) = (-1, 1, -2)$$

Form the matrix whose rows are the generators of U and row reduce to the echelon form

$$\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 1 & -2 \end{pmatrix} \xrightarrow[H_{31}(1)]{H_{21}(-2)} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{H_{32}(-1)} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus $\{(1, 0, 1), (0, 1, -1)\}$ is a basis of U , and so $\dim U = 2$

Linear Mappings

(ii) we seek the set of (x, y, z) such that $F(x, y, z) = (0, 0, 0)$, i.e.

$$F(x, y, z) = (x + 2y - z, y + z, x + y - 2z) = (0, 0, 0)$$

Set corresponding components equal to each other to form the following homogeneous system whose solution space is the kernel W of T :

$$\begin{array}{l} x + 2y - z = 0 \\ y + z = 0 \\ x + y - 2z = 0 \end{array} \quad \Rightarrow \quad \begin{array}{l} x + 2y - z = 0 \\ y + z = 0 \\ -y - z = 0 \end{array} \quad \Rightarrow \quad \begin{array}{l} x + 2y - z = 0 \\ -y - z = 0 \end{array}$$

The only free variable is z ; hence $\dim W = 1$. Let $z = 1$; then $y = -1$ and $x = 3$.

Thus $\{(3, -1, 1)\}$ is a basis of W .

5.34. Let U and W be the subspaces of $\mathbb{R}^4$ generated by

$$\{(1, 1, 0, -1), (1, 2, 3, 0), (2, 3, 3, -1)\} \quad \text{and} \quad \{(1, 2, 2, -2), (2, 3, 2, -3), (1, 3, 4, -3)\}$$

respectively. Find (i) $\dim(U + W)$, (ii) $\dim(U \cap W)$.

- (i) $U + W$ is the space spanned by all six vectors. Hence form the matrix whose rows are the given six vectors, and then row reduce to echelon form:

$$\begin{pmatrix} 1 & 1 & 0 & -1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & -1 \\ 1 & 2 & 2 & -2 \\ 2 & 3 & 2 & -3 \\ 1 & 3 & 4 & -3 \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 4 & -2 \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since the echelon matrix has three nonzero rows, $\dim(U + W) = 3$.

- (ii) First find $\dim U$ and $\dim W$. Form the 4×4 matrices whose rows are the generators of U and W respectively and then row reduce each to echelon form.

$$\begin{pmatrix} 1 & 1 & 0 & -1 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 3 & -1 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 1 & 3 & 1 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 2 & 2 & -2 \\ 2 & 3 & 2 & -3 \\ 1 & 3 & 4 & -3 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 2 & 2 & -2 \\ 0 & -1 & -2 & 1 \\ 0 & 1 & 2 & -1 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 2 & 2 & -2 \\ 0 & -1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since each of the echelon matrices has two nonzero rows, $\dim U = 2$ and $\dim W = 2$. Using Theorem 5.8 that $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$, we have

$$3 = 2 + 2 - \dim(U \cap W) \quad \text{or} \quad \dim(U \cap W) = 1$$

5.35. Let U be the subspace of $\mathbb{R}^6$ generated by

$$\{(1, 3, -2, 2, 3), (1, 4, -3, 4, 2), (2, 3, -1, -2, 9)\}$$

and let W be the subspace generated by

$$\{(1, 3, 0, 2, 1), (1, 5, -6, 6, 3), (2, 5, 3, 2, 1)\}$$

Find a basis and the dimension of (i) $U+W$, (ii) $U \cap W$.

- (i) $U+W$ is the space generated by all six vectors. Hence form the matrix whose rows are the six vectors and then row reduce to echelon form:

$$\begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 1 & 4 & -3 & 4 & 2 \\ 2 & 3 & -1 & -2 & 9 \\ 1 & 3 & 0 & 2 & 1 \\ 1 & 5 & -6 & 6 & 3 \\ 2 & 5 & 3 & 2 & 1 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & -3 & 3 & -6 & 3 \\ 0 & 0 & 2 & 0 & -2 \\ 0 & 2 & -4 & 4 & 0 \\ 0 & -1 & 7 & -2 & -5 \end{pmatrix}$$

$$\text{to } \begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & -2 \\ 0 & 0 & -2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 0 & 2 & 0 & -2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

The set of nonzero rows of the echelon matrix,

$$\{(1, 3, -2, 2, 3), (0, 1, -1, 2, -1), (0, 0, 2, 0, -2)\}$$

is a basis of $U+W$; thus $\dim(U+W) = 3$.

- (ii) First find homogeneous systems whose solution sets are U and W respectively. Form the matrix whose first rows are the generators of U and whose last row is (x, y, z, s, t) and then row reduce to echelon form:

$$\begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 1 & 4 & -3 & 4 & 2 \\ 2 & 3 & -1 & -2 & 9 \\ x & y & z & s & t \end{pmatrix} \text{ to } \begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & -3 & 3 & -6 & 3 \\ 0 & -3x+y & 2x+z & -2x+s & -3x+t \end{pmatrix}$$

$$\text{to } \begin{pmatrix} 1 & 3 & -2 & 2 & 3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 0 & -x+y+z & 4x-2y+s & -6x+y+t \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$